

A machine-checked proof of an Amos-type bound for modified Bessel ratios at real order

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Abstract

The Amos-type upper bound for the modified Bessel function ratio,

$$\rho_\nu(x) = \frac{I_{\nu+1}(x)}{I_\nu(x)} < \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2}} = B_\nu(x),$$

is classical, and its derivation through the qualitative theory of the associated Riccati equation is an established technique. A companion paper formalized the bound at integer order over a factorial power series. This paper extends the formalization to *every real order* $\nu \geq 0$: the function I_ν is defined by its Γ -power series (real exponents via `rpow`), and the complete chain — convergence, positivity, the three-term recurrence, termwise differentiation with a dominated-derivative argument that must treat the leading term separately (its exponent $\nu - 1$ is negative for $\nu < 1$), the Riccati equation, a small-argument zone bound uniform in ν , and a first-crossing barrier — is machine-checked in Lean 4 with axiom oracle `[propext, Classical.choice, Quot.sound]` and no analytic hypothesis beyond $\nu \geq 0$, $x > 0$. Two structural locks tie the result to the integer development: an *identification theorem* proves that at $\nu = n \in \mathbb{N}$ the Γ -series object coincides with the factorial-series object, so the integer-order theorem of the companion development is recovered as a corollary in three rewrites; and a genuinely non-integer instance at $\nu = \frac{1}{2}$ witnesses that the endpoint lives outside the natural-number embedding. The theorem is proved for the in-core Γ -series definition; no identification with an external special-functions library object is claimed.

1 Introduction

The ratio $\rho_\nu = I_{\nu+1}/I_\nu$ of modified Bessel functions of the first kind is bounded above by the algebraically calibrated expression $B_\nu(x) = x/(\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2})$; bounds of this shape trace back to Amos [1], and sharp forms and systematic families were developed by Segura [2], Hornik and Grün [3, 5], and Ruiz-Antolín and Segura [4], the latter describing explicitly the Riccati-based qualitative methodology that classical proofs of such bounds employ. *No novelty is claimed for the bound or for the Riccati approach.* We refer to the inequality as an *Amos-type* bound: the attribution of this exact display to a specific equation of [1] is deliberately not

*This manuscript and parts of the formal development were produced with AI assistance under the repository’s audit protocol. Every theorem below is machine-checked in Lean 4 against a pinned Mathlib; the proof-independent numerical cross-check of Section 6 is a committed interval-arithmetic transcript; remaining numerics are labelled paper-level where they occur. Repository: <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>.

asserted here pending collation with the primary source. The displayed upper bound is precisely the first inequality of Ruiz-Antolín and Segura [4, Theorem 2], with their order replaced by $\nu+1$. Their classical range becomes $\nu \geq -1/2$; the present formalization proves the restriction $\nu \geq 0$ and does not claim the remaining negative-order interval.

The contribution of this paper is the formalization at *real order*. Two companion papers of this series formalized, over the factorial power series at integer order, the consequence calculus of the bound [7] and then the bound itself [8]. The latter states its own scope limit: no extension to noninteger order is claimed. This paper removes that limit. The bound and its consumers below are stated for every real $\nu \geq 0$ and every $x > 0$, over the Γ -power series

$$\text{besselIReal } \nu \ x = \sum_{k \geq 0} \frac{(x/2)^{\nu+2k}}{k! \Gamma(\nu + k + 1)},$$

with the real exponent carried by `Real.rpow` and the denominator by `Real.Gamma`. In addition, the *identification theorem* `besselIReal_natCast` proves `besselIReal n = besselI n` at every integer order — so the integer development of [7, 8] is not a parallel theory but a proved section of the real one, and the integer-order bound of [8] becomes a corollary (Theorem 3.3).

Statuses: **exact** means a Lean theorem in the pinned central repository with clean axiom oracle; **certified** means a committed interval-arithmetic transcript; **paper-level** means ordinary prose, always flagged; *verified* marks an ordinary high-precision computation used only as a cross-check of a certified result, never as evidence. The repository’s largest namespace (`YangMills/`) reflects its original motivation in lattice gauge theory; no reduction to any continuum or gauge-theoretic problem is claimed or used.

2 Setting

All objects live in module `AmosClosure` of the pinned central repository. From the companion development [7, 8]: the factorial-series `besselI n x =  $\sum_{k \geq 0} (x/2)^{n+2k} / (k! (n+k)!)$` with its positivity, recurrence, derivative and ratio calculus, the predicate `AmosBound  $\nu \ x \ r \iff r < B_\nu(x)$` — already real in its order parameter — and the calibration engine. New in this paper, the real-order object `besselIReal  $\nu \ x$` above, its ratio $\rho_\nu = \text{besselRatioReal } \nu \ x$, and $\psi_\nu(x) = x(1/\rho_\nu - \rho_\nu)$ (`besselPsiReal`). Everything is stated at $x > 0$, the `rpow` domain of the development.

3 Main results

Theorem 3.1 (the Amos-type bound at real order; `amosBoundReal_holds`). *For every real $\nu \geq 0$ and every real $x > 0$,*

$$\frac{\text{besselIReal } (\nu+1) \ x}{\text{besselIReal } \nu \ x} < \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2}}.$$

No hypothesis remains beyond $\nu \geq 0$ and $x > 0$.

Theorem 3.2 (identification; `besselIReal_natCast`). *For every $n \in \mathbb{N}$ and $x > 0$,*

$$\text{besselIReal } (n : \mathbb{R}) \ x = \text{besselI } n \ x.$$

Termwise, `rpow` collapses to the natural power and $\Gamma(n + k + 1) = (n + k)!$. The integer-order development is a proved restriction of the real-order one.

Theorem 3.3 (consumers, recovery, and the half-order witness). *For every real $\nu \geq 0$ and $x > 0$: $\rho_\nu(x) - \rho_{\nu+1}(x) < 1/x$ (`besselIReal_unit_step`); the derivative of $\log \circ \text{besselIReal } \nu$ at x is strictly below that of $\log \circ \text{besselIReal } (\nu+1)$, in `deriv` form (`besselIReal_logDeriv_lt`). Moreover the integer-order theorem of [8] is recovered through Theorem 3.2 in three rewrites (`amosBound_holds_recovered`), and the instance $\nu = \frac{1}{2}$ is stated and proved as a named theorem (`amosBoundReal_half_order`) — the endpoint is not consumed only through the natural-number embedding.*

The φ -monotonicity step of [7] remains integer-indexed — its sequence is integer by construction and is already unconditional through [8]; no artificial real-order variant is introduced.

4 The proof

The formal proofs are the authority; this section presents the mathematics in full, at the level of the checklist a referee needs.

4.1 Series facts

Write $t_{\nu,k}(x) = (x/2)^{\nu+2k} / (k! \Gamma(\nu + k + 1))$. The exact term-ratio identity

$$t_{\nu,k+1} = t_{\nu,k} \cdot \frac{(x/2)^2}{(k+1)(\nu+k+1)} \quad (\text{besselRealTerm_succ})$$

follows from $\Gamma(s+1) = s \Gamma(s)$ at $s = \nu + k + 1 \geq 1$ and $(x/2)^{a+2} = (x/2)^a (x/2)^2$. Convergence: since $\Gamma(\nu+1) \leq \Gamma(\nu+k+1)$ for $\nu \geq 0$ (each functional-equation factor $\nu+j \geq 1$), the terms are dominated by an exponential series, and $I_\nu > 0$ follows termwise. The three-term recurrence $I_\nu - I_{\nu+2} = (2(\nu+1)/x) I_{\nu+1}$ is proved termwise (index shift plus the functional equation), exactly as at integer order. Theorem 3.2 is likewise termwise: $(x/2)^{(n:\mathbb{R})+2k} = (x/2)^{n+2k}$ by `Real.rpow_natCast` and $\Gamma(n+k+1) = (n+k)!$ by `Real.Gamma_nat_eq_factorial`.

4.2 The derivative layer

Termwise differentiation on the interval $(x/2, 3x/2) \subset (0, \infty)$ — the `rpow` domain forbids a ball through 0 — with dominated derivatives. The termwise derivative is

$$\frac{d}{dy} t_{\nu,k}(y) = \frac{\nu+2k}{2} \frac{(y/2)^{\nu+2k-1}}{k! \Gamma(\nu+k+1)}.$$

Remark 4.1 (the negative-exponent trap). *For $k = 0$ and $0 \leq \nu < 1$ the exponent $\nu - 1$ is negative: the factor $(y/2)^{\nu-1}$ is decreasing in y , and a majorant that evaluates it at the right endpoint is false. The formalized domination therefore treats $k = 0$ separately, with the two-endpoint bound $(y/2)^{\nu-1} \leq (x/4)^{\nu-1} + (3x/4)^{\nu-1}$ valid in both monotonicity regimes (at $\nu = 0$ the whole term carries the factor $\nu = 0$); and for $k \geq 1$ it uses the factoring $\nu+2k-1 = (\nu+1)+2(k-1)$, with $\nu+1 \geq 1$ and $2(k-1) \geq 0$, giving $(y/2)^{\nu+2k-1} \leq (3x/4)^{\nu+1} A^{k-1}$ with $A = (3x/4)^2$. The majorant tail is summable since $(\nu+2k+2)/(k+1)! \leq (\nu+2)/k!$, equivalent to $0 \leq \nu k$.*

Mathlib's dominated-differentiation theorem for series then yields, for every $\nu \geq 0$ and $x > 0$,

$$I'_\nu(x) = I_{\nu+1}(x) + \frac{\nu}{x} I_\nu(x) \quad (\text{besselIReal_hasDerivAt}),$$

with the `deriv`-form and logarithmic-derivative identities as corollaries, and the $\nu = n$ value test against the integer layer through Theorem 3.2 (`besselIReal\_deriv\_value\_natCast`).

4.3 The Riccati equation and the nullcline

From the derivative identity and the recurrence, the quotient rule gives

$$\rho'_\nu(x) = Q_x(\rho_\nu(x)), \quad Q_x(y) := 1 - \frac{2\nu+1}{x} y - y^2 \quad (\text{besselRatioReal_hasDerivAt}). \quad (1)$$

The bound B_ν satisfies the exact calibration identity

$$\frac{1}{B_\nu(x)} - B_\nu(x) = \frac{2\nu+1}{x} \quad (\text{amosRHS_calibration_real}), \quad (2)$$

with $a = \nu + \frac{1}{2} > 0$ supplied by $\nu \geq 0$; multiplied by $B_\nu > 0$ this is precisely $Q_x(B_\nu(x)) = 0$ (`riccatiQReal\_amosRHS`): the bound is the positive root of the Riccati quadratic. Moreover $Q_x(y) > 0$ for $0 \leq y < B_\nu(x)$, by the factorization $Q_x(y) - Q_x(B_\nu) = (B_\nu - y)(\frac{2\nu+1}{x} + y + B_\nu)$ (`riccatiQReal\_pos\_of\_lt`).

4.4 The zone: $x \leq 1/4$, uniformly in ν

The target $\rho_\nu < B_\nu$ is equivalent, by (2) and the strict antitonicity of $t \mapsto 1/t - t$ on $(0, \infty)$, to $\psi_\nu > 2\nu + 1$. On the zone this is proved directly from the series, with four product-form facts: geometric domination $t_{\nu,k} \leq t_{\nu,0} q_\nu^k$ with $q_\nu = (x/2)^2/(\nu+1)$, by induction from the term-ratio identity and $\nu+1 \leq (k+1)(\nu+k+1)$ (`besselRealTerm\_le\_geometric`); the product bound $(1 - q_\nu) I_\nu \leq t_{\nu,0}$ for $q_\nu < 1$ (`besselIReal\_mul\_le`); the strict two-term lower bound $t_{\nu,0} < I_\nu$ (`besselRealTerm\_zero\_lt\_besselIReal`) — the sole source of strictness in the whole chain; and the exact identity $2(\nu+1) t_{\nu+1,0} = x t_{\nu,0}$, derived by multiplying the already-proved termwise recurrence at $k = 0$ by x (`besselRealTerm\_zero\_succ`).

The ν -uniformity is isolated in two named lemmas rather than automation: $(\nu+1)/(\nu+2) \leq 1$ (`real\_zone\_ratio\_uniform`), whence $2(\nu+1) q_{\nu+1} = \frac{x^2}{2} \cdot \frac{\nu+1}{\nu+2} \leq \frac{x^2}{2}$, and with $x^2 \leq 1/16$ the coefficient inequality

$$2\nu + 1 + x^2 \leq 2(\nu+1)(1 - q_{\nu+1}) \quad (\text{real_zone_coefficient_bound}).$$

Chaining (upper bound at order $\nu+1$, identity, strict lower bound at order ν) gives the decisive strict inequality

$$(2\nu + 1 + x^2) I_{\nu+1}(x) < x I_\nu(x) \quad (0 < x \leq 1/4), \quad (3)$$

strictness entering only at the last link. Two further lines convert (3) into the zone claim: since $2\nu+1+x^2 \geq 1$, (3) also gives the crude bound $I_{\nu+1} \leq x I_\nu$, hence $x I_{\nu+1}^2 \leq x^2 I_{\nu+1} I_\nu$; multiplying (3) by $I_\nu > 0$ and inserting this,

$$(2\nu + 1) I_{\nu+1} I_\nu + x I_{\nu+1}^2 \leq (2\nu + 1) I_{\nu+1} I_\nu + x^2 I_{\nu+1} I_\nu < x I_\nu^2,$$

which, by the identity $\psi_\nu \cdot (I_{\nu+1} I_\nu) = x(I_\nu^2 - I_{\nu+1}^2)$, is precisely $\psi_\nu(x) > 2\nu + 1$ on the zone (`besselPsiReal\_zone`).

4.5 The barrier: every touch is an upward crossing

Suppose $\psi_\nu(c) = 2\nu + 1$ at some $c > 0$. Then $1/\rho_\nu(c) - \rho_\nu(c) = (2\nu + 1)/c$, and substituting into the Riccati quadratic annihilates it, so $\rho'_\nu(c) = 0$ (`riccati_zero_of_real_touch`); the product rule collapses $\psi'_\nu(c) = (2\nu + 1)/c > 0$, the numerator positive directly from $\nu \geq 0$. Every touch of the critical level is an upward crossing.

Lemma 4.2 (first crossing; the human form of `besselPsiReal_gt`). *Suppose, for contradiction, $\psi_\nu(x_1) \leq 2\nu + 1$ for some $x_1 > 0$. By the zone theorem $x_1 > 1/4$. The set $S = \{y \in [1/4, x_1] : \psi_\nu(y) \leq 2\nu + 1\}$ is nonempty, bounded below, and closed (continuity of ψ_ν on the interval, from differentiability); let $c = \inf S \in S$. Since $\psi_\nu(1/4) > 2\nu + 1$, in fact $c > 1/4$, and by minimality $\psi_\nu > 2\nu + 1$ on $(1/4, c)$ (at $1/4$ itself this holds by the zone theorem); left-continuity forces the touch $\psi_\nu(c) = 2\nu + 1$. But then $\psi'_\nu(c) = (2\nu + 1)/c > 0$, so the difference quotient $(\psi_\nu(y) - \psi_\nu(c))/(y - c)$ is eventually positive as $y \uparrow c$; since $y - c < 0$ there, $\psi_\nu(y) < \psi_\nu(c) = 2\nu + 1$ just left of c — contradicting $\psi_\nu > 2\nu + 1$ on $(1/4, c)$. Hence $\psi_\nu(x) > 2\nu + 1$ for all $x > 0$.*

Finally, $\psi_\nu > 2\nu + 1$ converts into $\rho_\nu < B_\nu$ algebraically: with $s = 1/\rho_\nu - \rho_\nu$ and $t = 1/B_\nu - B_\nu$,

$$(s - t) \cdot \rho_\nu B_\nu = (B_\nu - \rho_\nu) (1 + \rho_\nu B_\nu),$$

and the left-hand side is positive — both its factors are: $s > t$ is the barrier conclusion $\psi_\nu > 2\nu + 1$ divided by x , and $\rho_\nu B_\nu > 0$ — so, since $1 + \rho_\nu B_\nu > 0$, positivity passes to $B_\nu - \rho_\nu$; this is the closing step of `amosBoundReal_holds`. That completes Theorem 3.1; Theorem 3.3 is a direct application, the recovery corollary consisting of the two identification rewrites, one cast normalization, and `exact`.

5 Scope

The theorem is proved for the in-core Γ -power-series definition of the modified Bessel function at real order $\nu \geq 0$, $x > 0$. No formal identification with an external special-functions library object is claimed (none exists at the pinned Mathlib to identify with). The integer-order objects of the companion development are identified with the real-order object *by theorem* (Theorem 3.2) — the corresponding scope caveat of [8] is thereby discharged. Results elsewhere in this programme that consume Amos-type bounds for the classical I_ν of the analysis literature — such as the surface-track φ -lemma conditional discussed in [7] — do not change verification status through this paper.

6 Proof-independent numerical cross-check

A certified interval-arithmetic companion evaluates the margin — its protocol pre-registered in the development's charter (`docs/C4-CHARTER.md`) before the authoritative run — $\Delta(\nu, x) = B_\nu(x) - \rho_\nu(x)$ at all 70 points of the pre-registered grid $\nu \in \{0, \frac{1}{10}, \frac{1}{2}, \frac{9}{10}, 1, \frac{3}{2}, 2, \pi, 10, 100\}$, $x \in \{\frac{1}{8}, \frac{1}{4}, 1, 2, 5, 50, 300\}$, with π a certified ball. The ratio is computed by the certified backward recurrence $\rho_\alpha = x/(2(\alpha + 1) + x\rho_{\alpha+1})$, anchored at order $\nu + N$ with the anchor condition $q \leq \frac{1}{4}$ itself certified in ball arithmetic and seeded by the two-sided product-form bounds of Section 4.4 — no exponentially large value of I_ν is ever formed, and the seed is independent of Theorem 3.1. Verdicts are by ball sign (`PASS` iff $\inf \Delta > 0$), never by midpoint. All 70 points `PASS` at 128 bits; a methodologically separate ordinary high-precision pass (labelled *verified*, not certified) agrees at

all 70 points; and at $(\nu, x) = (0, 300)$ the certified margin $[2.78243072419 \times 10^{-6} \pm 2.43 \times 10^{-19}]$ reproduces, by this entirely different method, the slack floor measured by the integer-order companion of [7]. It is *not* part of the proof, and the proof does not cite it. Both computations are author-repository artifacts; methodological separation does not establish external operator or evaluator independence.

7 Related work

Amos [1] introduced bounds of this shape in the context of computing modified Bessel ratios; two-sided sharp families and Turán-type consequences are due to Segura [2]; Hornik–Grün studied the Amos-type family systematically [3] and proved in-family optimality results [5]; Ruiz-Antolín–Segura [4] give the sharp bounds at both parameter ends and describe the Riccati-based qualitative methodology to which the route formalized here belongs. The recurrence and derivative identities are [6, 10.29], and the Γ -series is the standard definition [6, 10.25.2]. No systematic priority survey is supplied, and no worldwide first- formalization claim is made. The integer-order formalization and its consequence calculus are in the companion papers [7, 8]; the prose companions of the consumer families are [9, 10, 11]. The development builds on Mathlib [12].

8 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

07642720480157414db592fa85b626dafb71355b

(lakefile and manifest agree); `tectonic 0.15.0`; `python-flint 0.9.0` for the certified companion.

Recorded build facts: at the final mathematics commit

56ff479b863d8585c41f6738b7bd18f80f2cbb16

`lake build AmosClosure` completed successfully (**8171 jobs**); the axiom oracle printed, for all **74** registered statements (the 34 of the `c3-v1.0` release registry plus the 40 of this paper), exactly

`[propext, Classical.choice, Quot.sound];`

the oracle log is committed under `scripts/` at the same commit. The full run history of this development — eight build transcripts including the vacuous first run and every failed attempt, each committed the day it was produced with its diagnosis in the commit message; the certified-grid transcripts (the first kept but superseded; the second the authoritative script-and-transcript pair); and six charter amendments recording every external audit round and every method deviation — is committed under `scripts/` and `docs/C4-CHARTER.md`; per the repository’s audit protocol it is archival material and plays no role in the mathematics above.

Headline-theorem-to-artifact map (all formal results are in module `AmosClosure`; every **exact** row carries a direct oracle witness, while the **certified** row is supported by its script and transcript; the 34 inherited statements are mapped in [7, 8]):

Result	Lean name	File	Status
Γ -series interface	summable_besselRealTerm etc.	BesselRealInterface.lean	exact
Thm. 3.2	besselIReal_natCast	BesselRealInterface.lean	exact
derivative identity	besselIReal_hasDerivAt	BesselRealDeriv.lean	exact
Eq. (1)	besselRatioReal_hasDerivAt	RiccatiReal.lean	exact
Eq. (2)	amosRHS_calibration_real	RiccatiReal.lean	exact
zone (§4.4)	besselPsiReal_zone	AmosBoundProofReal.lean	exact
uniformity lemmas	real_zone_ratio_uniform etc.	AmosBoundProofReal.lean	exact
Lemma 4.2	besselPsiReal_gt	AmosBarrierReal.lean	exact
Thm. 3.1	amosBoundReal_holds	AmosBarrierReal.lean	exact
consumers	besselIReal_unit_step, _logDeriv_lt	AmosBarrierReal.lean	exact
recovery lock	amosBound_holds_recovered	AmosBarrierReal.lean	exact
half-order witness	amosBoundReal_half_order	AmosBarrierReal.lean	exact
§6 grid	c4_amos_real_arb.py + transcript v2	scripts/	certified

The commands below use the release revision `c4-v1.0`. Its oracle contains 96 registered statements, including 22 later C5 entries; the 74-entry oracle and 8171-job transcript reported above belong to the earlier mathematics commit `56ff479b863d8585c41f6738b7bd18f80f2cbb16`. The C4 proof modules are unchanged between these revisions. The final command compiles the original release manuscript; AIRR v2 is supplied as a separate source and PDF.

```
git clone https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME
cd THE-ERIKSSON-PROGRAMME
git checkout --detach cd62e1225e59335df974949d5c58a455cb790192
lake exe cache get
lake build AmosClosure
lake env lean AmosClosure/Oracle.lean
python scripts/c4_amos_real_arb.py
tectonic -X compile papers/c4-amos-real/c4_amos_real.tex
```

To compile this separate candidate, run the following from its supplied source directory (not the historical repository manuscript directory):

```
tectonic -X compile paper.tex
python c4_amos_real_arb_strict.py
```

The strict numerical companion is a new, separately hashed script. It preserves the numerical algorithm, returns a nonzero exit status for any FAIL or UNDECIDED result, and compares each mpmath sign with its stored certified verdict; disagreement also fails the command. The historical script and transcripts are retained unchanged. The original manuscript is preserved at tag `c4-v1.0`, commit

`cd62e1225e59335df974949d5c58a455cb790192`

Its tag-scoped `RELEASE-MANIFEST.md` records the original PDF and source hashes; compare Git blobs at that commit, accounting for checkout line endings. This AIRR v2 candidate is a separate manuscript revision, not an update of that tag. Its source, PDF, predecessor, changelog and hashes accompany the private submission. No release or editorial acceptance of v2 is asserted.

Revision verification and disclosure. The September 2026 revision was prepared with GPT-6 Astra assistance in Codex in response to the private GPT-5.6 Sol assessment of AIRR v1. This third private candidate incorporates precise classical attribution, explicit reproduction commands and a strict numerical companion following a formal GPT-6 Astra High report recommending minor revision on candidate 2. Formal sources and mathematical statements remain unchanged. A fresh local project build at the pinned release completed successfully on 10 September 2026: 8174 jobs, followed by 96 oracle entries containing only `[propxt, Classical.choice, Quot.sound]`. The raw build log, oracle log, environment summary and their hashes are included in this candidate’s supplementary package. The build used the pinned downloaded Mathlib cache; it is operator-run evidence, not an independent referee execution or a rebuild of every Mathlib dependency. The historical derivative-source comment saying both factored exponents are at least one is imprecise: the even exponent can be zero. Remark 4.1 states the correct qualification; the code already handles this case. The immutable source tag is preserved with this comment erratum. Earlier numerical transcripts record 70 PASS, 0 FAIL, 0 UNDECIDED at 128 bits and a 70/70 mpmath sign comparison. Their provenance remains historical; any run of the new strict companion is recorded separately. Assessments of preceding candidates do not transfer to this changed PDF. Evaluation and editorial decision for this exact candidate remain pending.

9 Conclusion

An Amos-type bound for modified Bessel ratios is now a theorem of the pinned formal development at every real order $\nu \geq 0$, proved from the in-core Γ -power series using only machine-checked analytic library results, with no additional analytic hypothesis beyond $\nu \geq 0$ and $x > 0$, and with the integer-order development of the companion papers identified as a proved restriction and recovered as a corollary. Limitations: the in-core series definition only, and $\nu \geq 0$ (Section 5); the classical bound and the Riccati methodology are prior art — the contribution is the machine-checked formalization of this complete chain in the repository. No claim of world-wide formalization priority is made. Further directions include comparison with sharper lower bounds, formal identification with a general special-functions interface if one becomes available in Mathlib, and upstreaming the modified-Bessel API.

References

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